Pandas Documentation: Working with DataFrames

Source: https://pandas.pydata.org/docs/getting_started/intro_tutorials/01_table_oriented.html

I want to start using pandas

To load the pandas package and start working with it, import the package. The community-agreed alias for pandas is `pd`.

```
import pandas as pd
```

Pandas Data Table Representation

To manually store data in a table, create a `DataFrame`. When using a Python dictionary of lists, the dictionary keys will be used as column headers, and the values in each list as columns of the DataFrame.

```
df = pd.DataFrame(
    {
        "Name": [
            "Braund, Mr. Owen Harris",
            "Allen, Mr. William Henry",
            "Bonnell, Miss Elizabeth",
        ],
        "Age": [22, 35, 58],
        "Sex": ["male", "male", "female"],
    }
)
df
```

Output:

```
                       Name  Age     Sex
0   Braund, Mr. Owen Harris   22    male
1  Allen, Mr. William Henry   35    male
2   Bonnell, Miss Elizabeth   58  female
```

A `DataFrame` is a 2-dimensional data structure that can store data of different types (including characters, integers, floating point values, categorical data and more) in columns. It is similar to a spreadsheet, a SQL table, or the `data.frame` in R.

- The table has 3 columns, each with a column label: `Name`, `Age`, and `Sex`.
- The column `Name` consists of textual (string) data, `Age` is numeric, and `Sex` is textual.
- The index labels each row. By default, this is a sequence of integers starting at 0.

Each Column in a DataFrame is a Series

If you're interested in working with the data in a single column, selecting it returns a pandas `Series`.

```
df["Age"]
```

Output:

```
0    22
1    35
2    58
Name: Age, dtype: int64
```

When selecting a single column of a pandas DataFrame, the result is a pandas Series. To select the column, use the column label in square brackets `[]`.

If you are familiar with Python dictionaries, the selection of a single column is very similar to the selection of dictionary values based on the key.

You can create a Series from scratch as well:

```
ages = pd.Series([22, 35, 58], name="Age")
```

A pandas Series has no column labels, as it is just a single column of a DataFrame. A Series does have row labels.

Doing Something with a DataFrame or Series

To find the maximum Age of the passengers, select the `Age` column and apply `max()`:

```
df["Age"].max()
# 58
```

Or on the Series directly:

```
ages.max()
# 58
```

pandas provides many functionalities, each a *method* you can apply to a DataFrame or Series.

For basic statistics of the numerical data in the table, use `describe()`:

```
df.describe()
```

Output:

```
             Age
count   3.000000
mean   38.333333
std    18.230012
min    22.000000
25%    28.500000
50%    35.000000
75%    46.500000
max    58.000000
```

The `describe()` method provides a quick overview of the numerical data in a DataFrame. As the `Name` and `Sex` columns are textual data, these are by default not taken into account by `describe()`.

Many pandas operations return a DataFrame or a Series.

Remember

- Import the package: `import pandas as pd`.
- A table of data is stored as a pandas `DataFrame`.
- Each column in a `DataFrame` is a `Series`.
- You can do things by applying a method on a `DataFrame` or `Series`.